

K⁺/H⁺ Antiport in Heart Mitochondria*

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Heart mitochondria depleted of endogenous divalent cations by treatment with A23187 and EDTA swell in (a) K⁺ acetate or (b) K⁺ nitrate when an uncoupler is present. These mitochondria also exchange matrix ⁴²K⁺ with external K⁺, Na⁺, or Li⁺ in a reaction that does not require respiration and is insensitive to uncouplers. Untreated control mitochondria do not swell in either medium nor do they show the passive cation exchange. Both the swelling and the exchange reactions are inhibited by Mg²⁺ and by quinine and other lipophilic amines. Swelling and exchange are both strongly activated at alkaline pH, and the exchange reaction is also increased markedly by hypotonic conditions. All of these properties correspond to those reported for a respiration-dependent extrusion of K⁺ from Mg²⁺-depleted mitochondria, a reaction attributed to a latent Mg²⁺- and H⁺-sensitive K⁺/H⁺ antiport. The swelling reactions are strongly inhibited by dicyclohexylcarbodiimide reacted under hypotonic conditions, but the exchange reaction is not sensitive to this reagent. Heart mitochondria depleted of Mg²⁺ show marked increases in their permeability to H⁺, to anions, and possibly to cations, and the permeability to each of these components is further increased at alkaline pH. This generalized increase in membrane permeability makes it likely that K⁺/H⁺ antiport is not the only pathway available for K⁺ movement in these mitochondria. It is concluded that the swelling, ⁴²K⁺ exchange, and K⁺ extrusion data are all consistent with the presence of the putative K⁺/H⁺ antiport but that definitive evidence for the participation of such a component in these reactions is still lacking.

Cation/H⁺ antiports capable of extruding excess cation from the mitochondrial matrix are an integral part of the chemiosmotic coupling model (1, 2). Considerable experimental evidence (reviewed in Refs. 3 and 4) supports the existence of a Na⁺/H⁺ exchange component. There are indications that a separate K⁺/H⁺ antiport is also present, but the picture is by no means as clear. Several studies (5-9) suggest that the latter activity may be latent and regulated in such a way as to maintain a low-exchange activity under normal circumstances. Garlid (9) has extended this concept to a plausible model in which matrix Mg²⁺ binds reversibly to a regulatory site on the K⁺/H⁺ antiport to prevent its activity (the "Mg²⁺ carrier brake"). This model is supported by the finding that mitochondria, depleted of divalent cations by the ionophore A23187, extrude endogenous K⁺ in a respiration-dependent reaction that has properties consistent with K⁺/H⁺ antiport

(6-9). This extrusion of K⁺ is increased at elevated pH and in hypotonic media (6), sensitive to uncouplers and inhibitors of respiration (8), and inhibited by quinine (10).

Depletion of mitochondrial divalent cations with A23187 and EDTA has also been shown to induce passive osmotic swelling of rat liver mitochondria suspended in K⁺ nitrate following addition of an uncoupler (5), a result consistent with K⁺ uptake by K⁺/H⁺ antiport (see Refs. 3, 4, and 11). The passive uptake of K⁺ and osmotic swelling observed under these conditions probably has little physiological relevance, but such swelling reactions offer the possibility of a convenient assay system that can be used to establish the properties of the putative K⁺/H⁺ antiport.

Along these lines, Nakashima and Garlid (10) have recently used the swelling of divalent cation-depleted liver mitochondria in K⁺ acetate as an indication of K⁺/H⁺ antiport activity. These studies led them to conclude that the Na⁺/H⁺ antiport is highly specific for Na⁺ and insensitive to quinine or Mg²⁺, whereas the K⁺/H⁺ exchanger transports both Na⁺ and K⁺, is regulated by Mg²⁺, and is inhibited by H⁺ and by quinine (10). More recently this group has used the acetate swelling assay to establish that the K⁺/H⁺ antiport is strongly and irreversibly inhibited by DCCD¹ and that the inhibition is prevented by Mg²⁺, by quinine, and by low pH (12). DCCD labels an 82,000-dalton protein when it inactivates K⁺/H⁺ antiport in liver mitochondria (12). These important conclusions all depend on the assumption that the assay employed (swelling in acetate salts at elevated pH) can distinguish between cation movements by K⁺/H⁺ exchange and those promoted by membrane alterations that increase uniport conductivity (Ref. 13, for example).

The present study examines the swelling of beef heart mitochondria under conditions in which K⁺ can be presumed to be taken up by the putative K⁺/H⁺ antiport reaction. The properties of the passive osmotic swelling reactions are compared with the exchange and loss of matrix ⁴²K⁺ in Mg²⁺-depleted heart mitochondria. Both the swelling and exchange reactions are activated by Mg²⁺ depletion, by elevated pH, and by hypotonic conditions, and inhibited by exogenous Mg²⁺ and by quinine and quinacrine. These properties correspond to those of the K⁺/H⁺ antiport as defined by the respiration-dependent extrusion of K⁺ (6-9). However, the conditions necessary for optimum rates of swelling lead to greatly increased permeability to H⁺, to anions, and quite likely to K⁺ as well. This, along with the lack of specificity of the available inhibitors, makes the pathway for K⁺ flux somewhat uncertain. It is concluded that the swelling, ⁴²K⁺ exchange, and K⁺ extrusion data are all consistent with the presence of the putative Mg²⁺- and H⁺-sensitive K⁺/H⁺ antiport (9, 10, 12)

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¹ Abbreviations used are as follows: DCCD, dicyclohexylcarbodiimide; CCCP, carbonyl cyanide *m*-chlorophenylhydrazide; SMP, sub-mitochondrial particles; TEA⁺, tetraethylammonium ion; TMA⁺, tetramethylammonium ion; BSA, bovine serum albumin; TES, *N*-[tris(hydroxymethyl)methyl-2-amino]ethanesulfonic acid.

but that definitive evidence for the participation of such a component in these reactions is still lacking.

MATERIALS AND METHODS²

RESULTS AND DISCUSSION

Swelling of Heart Mitochondria in K⁺ Nitrate—Heart mitochondria, like those from liver (5), swell passively in K⁺ nitrate when (a) A23187 and EDTA are added to deplete endogenous Mg²⁺ and (b) an uncoupler (CCCP in Fig. 1) is added to confer permeability to H⁺. There is no swelling unless divalent cations are depleted (not shown) and no swelling in the absence of an uncoupler (Fig. 1). The maximum rate of swelling is low in these mitochondria and is attained only after an extended lag period (Fig. 1).

Heart mitochondria depleted of divalent cations and reisolated prior to the swelling assay (Preparation I in Fig. 1) also require addition of an uncoupler to swell in K⁺ nitrate. Again, the maximum rate of swelling is attained only after a lag period, and since the divalent cations have been removed in a prior step, the lag cannot be associated with the time required for Mg²⁺ or Ca²⁺ depletion. Swelling of divalent cation-depleted mitochondria in K⁺ nitrate following addition of an uncoupler is inhibited by the addition of Mg²⁺ and by prior reaction of the depleted mitochondria with DCCD (Fig. 1). The swelling is also inhibited by quinine (10) added either prior to CCCP or after the swelling reaction has attained its maximum velocity (Fig. 1).

The maximum rate of swelling of divalent cation-depleted mitochondria in K⁺ nitrate increases with increasing concentrations of CCCP in the range from 0.01 to 50 μ M with half-maximal activation near 0.1 μ M CCCP. Higher concentrations of CCCP (100 μ M) do not produce higher rates of swelling (records not shown). These results are consistent with a requirement for a saturating level of uncoupler to provide a pathway for H⁺ conductance and the subsequent entry of K⁺ by an obligatory K⁺/H⁺ exchange process (see Fig. 2). At less than saturating concentrations of uncoupler, the rate of swelling is a function of the CCCP level, and swelling appears to be limited by the rate of H⁺ penetration.

Heart mitochondria depleted of Mg²⁺ in a KCl medium retain high concentrations of K⁺ (Table I), and it is possible that the low rate of swelling of these mitochondria in K⁺ nitrate (Fig. 1) is related to the low concentration gradient for K⁺ under these conditions. Accordingly, heart mitochondria were depleted of K⁺, as well as divalent cations, by treatment with A23187 in a TEA⁺ medium (Preparation II in Fig. 1). This procedure is analogous to that used by Nakashima and Garlid (10) for liver mitochondria. These cation-depleted heart mitochondria swell at a significant rate in K⁺ nitrate in the absence of an uncoupler (Fig. 1). Swelling is stimulated still further by addition of CCCP, however, and the response to uncoupler shows no lag period (Fig. 1). These results suggest that a greater concentration gradient for K⁺ may increase the rate of passive osmotic swelling of Mg²⁺-depleted mitochondria in K⁺ nitrate. However, the partial loss of the uncoupler dependence of swelling also suggests

that the mitochondria have become more permeable to H⁺ when matrix K⁺ is replaced by TEA⁺ (see Fig. 2). Alternatively, a portion of the swelling in these mitochondria could result from K⁺ entry via a uniport pathway (Fig. 2).

Swelling in K⁺ Acetate—Untreated heart mitochondria swell only slowly when suspended in 100 mM K⁺ acetate (11, 18), and this has been taken as an indication that K⁺/H⁺ antiport activity is low in these preparations. Significant rates of swelling can be elicited in this medium, however, when divalent cations are depleted by addition of A23187 and EDTA. As with the swelling in K⁺ nitrate, the maximum rate of swelling is low under these conditions and is attained only after a lag period that exceeds the time necessary to deplete Mg²⁺. Mitochondria depleted of divalent cations by prior incubation in a K⁺ medium swell only slowly in H⁺ acetate, whereas those depleted in TEA⁺ swell much more extensively and at more rapid rates (Fig. 3A). However, mitochondria depleted of divalent cations and K⁺ in a sucrose medium (Fig. 3A) or in a medium of NaCl or trimethylammonium chloride (not shown) show rates of swelling in K⁺ acetate that are nearly identical to those of mitochondria depleted in KCl (Fig. 3A). This suggests that factors other than simply the divalent cation content of the mitochondria and the existing K⁺ concentration gradients are contributing to the observed swelling process.

Heart mitochondria depleted of divalent cations in the KCl medium swell at rates only slightly greater than control when suspended in 0.1 M ammonium nitrate (Fig. 3B), whereas those depleted in TEA⁺ show rapid and extensive swelling in this medium (Fig. 3B). As will be discussed in the following section, this result can be taken as an indication that the TEA⁺-treated mitochondria are more permeable to H⁺ (and/or NH₄⁺) than those depleted in other media.

The high rates of swelling in K⁺ acetate seen with mitochondria depleted of divalent cations and K⁺ in TEA⁺ increase still further as the pH of the medium is increased (Fig. 4). A similar pH response is seen when mitochondria are depleted of Mg²⁺ and K⁺ in a TMA⁺ medium (not shown). However, mitochondria depleted to similar low levels of Mg²⁺ and K⁺ in a sucrose medium show only low rates of swelling throughout this pH range (Fig. 4). Similar low rates of swelling in K⁺ acetate are seen when mitochondria are depleted of Mg²⁺ and K⁺ (to levels comparable to those shown in Table I) in a NaCl or a trimethylammonium chloride medium (data not shown). It is concluded that the rate of swelling of Mg²⁺-depleted heart mitochondria in both K⁺ nitrate (Fig. 1) and K⁺ acetate (Figs. 3 and 4) is strongly dependent on the conditions used to remove the divalent cations.³

The effect of various inhibitors on the swelling of heart mitochondria in K⁺ acetate was assessed using a pH of 7.8 and mitochondria depleted of both Mg²⁺ and K⁺ in a TEA⁺ medium (other conditions as for Fig. 4). Under these conditions, swelling is inhibited by Mg²⁺ and Ca²⁺ (I₅₀ of 2.5 and 0.3 mM, respectively), by quinine (I₅₀ of 0.4 mM), quinacrine (I₅₀ of 0.1 mM), and by nupercaine (I₅₀ of 0.5 mM). Swelling under these conditions is not sensitive to butacaine (0.5 mM), to CCCP (1.5 μ M), or to ADP (1 mM) and oligomycin (2 μ g·mg⁻¹ protein). In agreement with the findings of Nakashima and Garlid (10) using liver mitochondria, the swelling of

² Portions of this paper (including "Materials and Methods," Tables I-III, and Figs. 1-10) are presented in miniprint at the end of this paper. Miniprint is easily read with the aid of a standard magnifying glass. Full size photocopies are available from the Journal of Biological Chemistry, 9650 Rockville Pike, Bethesda, MD 20814. Request Document No. 84M-1559, cite the authors, and include a check or money order for \$4.80 per set of photocopies. Full size photocopies are also included in the microfilm edition of the Journal that is available from Waverly Press.

³ The reason for this differential response is not clear. Some possible explanations include: (a) differences in matrix pH (although studies of the distribution of [¹⁴C]acetate do not show marked differences between the various preparations), (b) small differences in matrix volume (again, estimates of matrix water content do not show significant differences), and (c) possible direct effects of the various solutes on membrane permeability and on matrix osmotic pressure.

cation-depleted heart mitochondria in Na⁺ acetate (100 mM, other conditions as for Fig. 1) increases at elevated pH, and the pH-dependent increment in swelling in this salt is inhibited by quinine (0.5 mM, data not shown).

Swelling in Ammonium Nitrate—Untreated heart mitochondria suspended in NH₄⁺ nitrate do not swell until an uncoupler is added to provide an influx of H⁺ (24, 25). The buildup of an ion pair to produce osmotic swelling of the matrix appears to require entry of H⁺ as well as the permeant NH₄⁺ free base and the nitrate anion (24). Swelling of uncoupled mitochondria increases with increased pH in this medium (24), a result that is consistent with the presence of a pH-sensitive anion pore (26). Swelling rates estimated for untreated heart mitochondria reflect these properties (Table II).

Heart mitochondria depleted of K⁺ and divalent cations in TEA⁺ show much higher rates of swelling in the absence of CCCP at both pH 7.1 and 8.2 (Table II). This suggests that the permeability of the mitochondria to H⁺ (or to NH₄⁺) has increased as a result of the depletion of internal cations. Very high rates of swelling are produced when CCCP is also added under these conditions. Since H⁺ permeation is not limiting in the presence of CCCP, the 6-fold increase in swelling rate seen in cation-depleted relative to untreated mitochondria (Table II) must reflect an increase in permeability to nitrate as a result of cation depletion (or the manipulations necessary for the cation-depletion process). A further 3-fold increase in the rate of swelling in the presence of CCCP is seen when the pH is increased to 8.2 (Table II). This suggests that the putative pH-sensitive anion pore (26) is also present and functioning and provides a pathway for anion penetration that is additive to that opened by the depletion and divalent cations. These results indicate that both the permeability to H⁺ (or NH₄⁺) and permeability to anions is increased by divalent cation depletion and that both permeabilities are further increased at alkaline pH. Beavis and Garlid (27) have reached a similar conclusion from a survey of swelling of liver mitochondria in these media.

Swelling of cation-depleted heart mitochondria in K⁺ nitrate is stimulated by CCCP at neutral pH as discussed above, but the higher rates of swelling in this medium at pH 8.2 are virtually insensitive to the uncoupler (Table II). It is possible that swelling under these conditions still depends on K⁺/H⁺ antiport but that the cation-depleted mitochondria have developed sufficient permeability to H⁺ so that the increased H⁺ conductivity of the uncoupler is not necessary (see Fig. 2). If this were the case, no response to CCCP would be seen. However, there is no evidence that K⁺ entry occurs solely via the putative antiport under these conditions. Since H⁺ and anion permeability are both increased (as indicated by the swelling in NH₄⁺ nitrate), the results clearly suggest that a generalized loss of membrane barrier properties has been produced.

In the presence of saturating levels of valinomycin, the rate of swelling of cation-depleted heart mitochondria in K⁺ nitrate is increased with increasing pH, another indication that anion permeability is increased under these conditions (Table II). The response to valinomycin establishes that the cation-depleted heart mitochondria are not freely permeable to K⁺ even at pH 8.2 and that the addition of an exogenous K⁺ uniport will still increase the rate of swelling in these mitochondria (see Fig. 2).

Swelling of Mg²⁺- and K⁺-depleted heart mitochondria in NH₄⁺ nitrate is inhibited by Mg²⁺ (2.5 mM inhibits 90%) and by quinine (0.5 mM inhibits 50%, data not shown). Since swelling in NH₄⁺ nitrate presumably does not involve the K⁺/H⁺ antiport, this result establishes that neither Mg²⁺ nor

quinine is absolutely specific for inhibition of the antiport reaction.

Passive Exchange of Matrix ⁴²K⁺ with External Cations—Heart mitochondria depleted of divalent cations lose matrix ⁴²K⁺ rapidly to a KCl, NaCl, or LiCl medium (8, 23). Since there is little loss of matrix ⁴²K⁺ to choline chloride (23) and the mitochondria retain high levels of matrix K⁺ at reduced specific activity (23), the loss of ⁴²K⁺ to media containing alkali metal salts can be considered to be a ⁴²K⁺/cation exchange reaction. The exchange does not require respiration, is insensitive to uncouplers, and increases with increasing pH (23). The reaction is strongly inhibited by Mg²⁺, Ca²⁺, and Mn²⁺ and does not occur unless the endogenous divalent cations are depleted (8, 23). Since the passive K⁺/cation⁺ exchange develops under the same conditions as the respiration-dependent extrusion of K⁺ attributed to K⁺/H⁺ antiport (7, 8), it seems likely that the exchange could also take place on the "unbraked" antiporter that is operating in the absence of a metabolic ΔpH (see Fig. 2). If this is the case, a study of the passive K⁺/cation⁺ exchange reaction should be the most direct way to obtain information on the properties of the antiporter.

Divalent cation-depleted mitochondria lose matrix ⁴²K⁺ rapidly to a KCl medium at pH 7.8 (Fig. 5). The loss of label is nearly linear over a 3-min incubation period and greatly exceeds the loss to a sucrose medium under the same conditions (Fig. 5). The loss of matrix ⁴²K⁺ increases with increasing external K⁺ at neutral pH when constant osmolality is maintained (Fig. 6). The loss of matrix ⁴²K⁺ to external K⁺, as well as the loss to sucrose, increases with increasing pH in the range from 7.0 to pH 8.2 (Fig. 6). The K⁺-dependent increment in ⁴²K⁺ loss is maximum at about pH 7.8, since at higher pH values the nonspecific loss to sucrose increases markedly (Fig. 6). The reaction in 100 mM KCl is strongly inhibited by quinine (1.0 mM) throughout this pH range (Fig. 6).

The passive loss of matrix ⁴²K⁺ to a KCl medium also increases when the osmolality of the suspending medium is decreased (Fig. 7). The loss of ⁴²K⁺ to KCl exceeds that to sucrose when osmolality decreases from 200 to 120 mosm, but further decreases in tonicity show no preference for KCl (Fig. 7). Matrix K⁺ values (estimated using tritiated water and [¹⁴C]sucrose, Ref. 28) show some decline indicative of a net loss of K⁺ under these hypotonic conditions (130 and 106 mM for no incubation and 3-min incubation, respectively, at pH 7.0 in 50 mM KCl at 25 °C) whereas the specific activity decreases from 60 to 5 cpm·nmol⁻¹ of K⁺. This establishes that the bulk of the ⁴²K⁺ loss is ⁴²K⁺/K⁺ exchange rather than net loss of matrix ⁴²K⁺. In the study shown in Fig. 8, virtually all matrix ⁴²K⁺ is lost to a suspending medium of 50 mM KCl (pH 7.0 to 8.2) in 3 min. This extensive loss of label is strongly inhibited throughout the pH range by quinine (Fig. 8). These studies of passive ⁴²K⁺/K⁺ exchange strongly support the conclusions of Bernardi and Azzone (6) that membrane stretching, as well as Mg²⁺ depletion and elevated pH, causes activation of K⁺/H⁺ antiport. The loss of matrix ⁴²K⁺ to hypotonic sucrose also becomes extensive in these mitochondria as the pH is increased from 7.0 to 8.2, and the loss of ⁴²K⁺ under these conditions is partially sensitive to quinine (Fig. 8).

The loss of matrix ⁴²K⁺ to sucrose above pH 8.0 (Figs. 6 and 8) could still reflect K⁺/H⁺ antiport if the primary effect of elevated pH is an increase in H⁺ permeability, so that net efflux of K⁺ via the antiport can occur without creating a limiting pH gradient (see Fig. 2). The sensitivity to quinine (Fig. 8) is the only criterion available to distinguish between

K⁺ loss by K⁺/H⁺ antiport, by K⁺ uniport, or by nonspecific membrane changes. For this reason it is uncertain whether increments in apparent ⁴²K⁺/K⁺ exchange above pH 8.0 indeed reflect increased antiporter activity.

Martin and Garlid (29) have interpreted the increased swelling of liver mitochondria in K⁺ acetate that is seen at elevated pH in terms of the removal of an inhibitory H⁺ from the K⁺/H⁺ antiporter. The present studies are compatible with this interpretation but suggest that other factors may also contribute to increased ion flux above about pH 8.0. Diwan (30) regards the pH dependence of K⁺ efflux from mitochondria as inconsistent with a passive K⁺/H⁺ antiport mechanism. However, the observation from the present work that passive ⁴²K⁺/K⁺ exchange increases with increasing pH (Fig. 6) suggests that increased activity at elevated pH may be a property of the antiporter and not of the availability of external H⁺.

The passive exchange of matrix ⁴²K⁺ with external KCl in divalent cation-depleted heart mitochondria shows a high temperature dependence that is consistent with a mediated exchange process (92% of matrix ⁴²K⁺ is lost to 100 mM KCl, pH 7.8, at 21 °C versus 26% for these condition at 4 °C). As was noted previously (23), the exchange process is not specific for external K⁺. The titrations shown in Fig. 9 establish that Na⁺ is slightly more effective than K⁺ or Li⁺ when compared at pH 7.8 in a medium containing 100 mM total salt (cation concentration adjusted with choline chloride). The pH of 7.8 was chosen for these surveys as a value that permits high K⁺/cation exchange with minimal loss to sucrose (see Fig. 6). These findings are in line with swelling studies (10) that indicate that the K⁺/H⁺ antiporter is not specific for K⁺. Passive exchange of mitochondrial K⁺ for Na⁺ is also seen in heart mitochondria swollen passively in nitrate salts (31), and these mitochondria also have lost a portion of their endogenous Mg²⁺. It should be noted that the exchange reaction does not appear to be saturated with respect to external monovalent cations under the conditions of Fig. 9. This may reflect a high degree of discrimination for H⁺ over monovalent cations on the external aspect of the antiporter in the pH range of these measurements.

The passive exchange of ⁴²K⁺ against 100 mM KCl (pH 7.8) is inhibited by quinine with an I₅₀ of about 0.4 mM. This value decreases to about 0.3 mM in 50 mM KCl at pH 7.0. Quinacrine is also an effective inhibitor of the reaction with I₅₀ values similar to quinine under both of these conditions. In 50 mM KCl, pH 7.0, nupercaine inhibits the passive ⁴²K⁺/K⁺ exchange (I₅₀ of 0.5 mM), as do Mg²⁺ and Ca²⁺ (I₅₀ of 2.5 and 0.2 mM, respectively). The exchange reaction in 100 mM KCl, pH 7.8, is somewhat more sensitive to Mg²⁺ and Ca²⁺ as indicated by lower I₅₀ values.

Effects of DCCD on K⁺/H⁺ Antiport in Heart Mitochondria—It has recently been reported that reaction of DCCD with Mg²⁺-depleted liver mitochondria at pH 7.8 results in a nearly complete and irreversible inhibition of K⁺/H⁺ antiport as measured using the swelling assay in K⁺ acetate (12). This inhibition was not seen in the presence of endogenous levels of Mg²⁺, at neutral pH, or in the presence of quinine, and this differential inhibition was used to identify an 82,000-dalton protein labeled with [¹⁴C]DCCD that is presumed to represent the antiport (12).

Swelling of heart mitochondria in K⁺ nitrate is sensitive to DCCD (Fig. 1), but the identical response is obtained when DCCD is reacted with untreated mitochondria (which are then Mg²⁺ depleted) or with previously Mg²⁺-depleted mitochondria. The presence of endogenous Mg²⁺ does not appear to protect the K⁺/H⁺ antiport of heart mitochondria from

inactivation by DCCD under these conditions (see Ref. 12).

The passive exchange of matrix ⁴²K⁺ from heart mitochondria is inhibited by only about 10–15% when the DCCD is reacted in the medium used to equilibrate the ⁴²K⁺ with the mitochondria. Martin *et al.* (12) reacted liver mitochondria with DCCD under hypotonic conditions (50 mM sucrose) to obtain complete inhibition of swelling of liver mitochondria in K⁺ acetate. Reacting heart mitochondria under hypotonic conditions (25 mM KCl) results in extensive inhibition of swelling in K⁺ acetate (Table III). A similar strong inhibition is obtained in 50 mM sucrose and 25 mM TEA⁺ chloride (not shown). The cation-depleted heart mitochondria that have been reacted with DCCD under conditions that produce extensive inhibition of swelling in K⁺ acetate still show vigorous ⁴²K⁺/K⁺ exchange (Table III). This occurs both in hypotonic KCl at neutral pH (Table III) and in 100 mM KCl at pH 7.8 (not shown). The exchange of ⁴²K⁺ with external cation remains sensitive to quinine so there has been no alteration of this property as a result of the extensive time of hypotonic incubation.⁴

These studies establish that DCCD must be reacted under hypotonic conditions in order to produce the inhibition of swelling in K⁺ acetate reported by Martin *et al.* (12) and that DCCD-reacted mitochondria (with inhibited acetate swelling) are still capable of extensive ⁴²K⁺/cation exchange. Two explanations for this set of results seem to be possible. (a) DCCD reacts with and inhibits the K⁺/H⁺ antiport (see Ref. 12). If this is correct then it seems likely that ⁴²K⁺/K⁺ exchange occurs by another mechanism. This in turn requires that an alternate pathway for K⁺ movement be postulated. This pathway would have to be available when Mg²⁺ is depleted and to be sensitive to exogenous Mg²⁺ and to quinine. (b) DCCD does not react with the K⁺/H⁺ antiport but inhibits swelling in K⁺ acetate by a mechanism that is not related to K⁺/H⁺ exchange. Such an explanation would call into question the relationship of the 82,000-dalton protein (12) to K⁺/H⁺ antiport. DCCD, like Mg²⁺ and quinine, also inhibits swelling of these mitochondria in NH₄⁺ nitrate (Table III), a reaction that is presumed not to depend on K⁺/H⁺ antiport.

Cation⁺/H⁺ Antiport in SMP—Na⁺- and Li⁺-dependent changes in the steady-state ΔpH of energized SMP have been detected using a quinacrine fluorescence assay (16). Similar protocols have been used to show K⁺-dependent H⁺ movements in *Escherichia coli* that are consistent with a K⁺/H⁺ antiport (32). Rosen and Futai (16) failed to detect K⁺/H⁺ antiport in SMP from liver mitochondria, but the presence of high levels of Mg²⁺ under the conditions they employed would presumably have prevented any such activity (9, 22). SMP from beef heart mitochondria show cation-dependent increases in ΔpH consistent with the presence of Na⁺/H⁺ antiport (Fig. 10) and Li⁺/H⁺ antiport (not shown) when respiring with succinate in the presence of EDTA. The K_m for the Na⁺-dependent response is 20 mM and that for Li⁺ is 2 mM, results that agree well with those for SMP from liver (16). These heart SMP show no response to added K⁺ (50 or 100 mM), but the addition of nigericin discharges ΔpH in the presence of K⁺ (Fig. 10).

The fluorescence assays were carried out using divalent cation-depleted particles at pH 8.0 in the presence of EDTA, conditions that should produce near maximum K⁺/H⁺ antiport activity based on both swelling and ⁴²K⁺/K⁺ exchange

⁴ Higher concentrations of DCCD result in a rapid loss of endogenous K⁺ indicative of nonspecific membrane alterations. For this reason it is necessary to incubate the mitochondria with no more than 50 nmol of DCCD/mg of protein and to extend the time of incubation.

studies. The Na⁺-dependent H⁺ movement in SMP (Fig. 10) provides powerful evidence for an obligatory link between Na⁺ and H⁺ flux or Na⁺/H⁺ antiport. The failure to detect K⁺-dependent H⁺ shifts leaves the K⁺/H⁺ antiport concept without such experimental support.⁵

Can Passive Swelling and Exchange in K⁺ Salts Be Attributed to a Latent K⁺/H⁺ Antiport?—Previous studies of the respiration-dependent extrusion of endogenous K⁺ from mitochondria (6–8) have concluded that this reaction can be attributed to a latent K⁺/H⁺ antiport. The putative antiport appears to be activated by depletion of endogenous Mg²⁺ (6–8, 22) and increased by elevated pH and decreased osmolarity of the suspending medium (6). The K⁺/H⁺ antiport also is inhibited by exogenous Mg²⁺ and Ca²⁺ (8) and by quinine and other amines (10). The present work has established that passive swelling in K⁺ acetate and K⁺ nitrate is activated and inhibited under the same conditions that activate and inhibit the respiration-dependent K⁺ extrusion. It, therefore, seems reasonable to conclude that these swelling reactions could depend on K⁺ entry into the matrix in exchange for internal H⁺ on the putative K⁺/H⁺ antiporter (see Fig. 2). In addition, a passive exchange of matrix ⁴²K⁺ for external cations is activated by depletion of matrix Mg²⁺, increased at alkaline pH, and inhibited by the same inhibitors as the respiration-dependent H⁺ extrusion. This activity can be viewed as an expression of the K⁺/H⁺ antiport operating in the absence of a metabolic ΔpH. The studies of K⁺ extrusion, swelling, and ⁴²K⁺/cation exchange all present a picture consistent with the presence of a latent Mg²⁺- and H⁺-sensitive K⁺/H⁺ antiport that is inhibited by Ca²⁺ and by quinine and other amines.

Despite the consistency of these data with the K⁺/H⁺ antiport postulates and the usefulness of this concept in explaining a number of observed phenomena, it is well to point out that definitive evidence for the existence of such a component is still lacking. The following considerations suggest that alternative explanations for the phenomena attributed to the K⁺/H⁺ antiport should be kept in mind.

⁵ It should be noted that a respiration-dependent uptake of K⁺ in SMP has been reported (33, 34) and that this uptake is stimulated by nigericin, an exogenous K⁺/H⁺ antiporter. The uptake of K⁺ was accompanied by net H⁺ uptake in these experiments, however, and K⁺-dependent H⁺ counter movements were not reported (33). Several explanations for the failure to detect K⁺-dependent H⁺ shifts in SMP (Ref. 33, Fig. 10) are available. The first (and most obvious) would be that there is no discrete K⁺/H⁺ antiporter in the mitochondrial membrane and that phenomena previously attributed to this component have other explanations. Arguing against this possibility is the observation that quinacrine is an effective inhibitor of swelling reactions attributed to K⁺/H⁺ antiport and passive ⁴²K⁺/K⁺ exchange. It is, therefore, possible that accumulation of the indicator during the fluorescence assay (Fig. 10) is sufficient to abolish K⁺/H⁺ activity. It should be noted that mitochondrial Na⁺/H⁺ and Li⁺/H⁺ antiport (16) and the K⁺/H⁺ antiport of *E. coli* (32) do not appear to be quinacrine sensitive, and this would require that the putative mitochondrial K⁺/H⁺ antiport differ markedly from these components in its response to quinacrine. Another possibility is that the antiporter (or an essential portion of it) is lost during the process of preparing SMP from intact mitochondria. Such a loss of K⁺/H⁺ antiport activity on conversion of one SMP (A particles) to another (ASU particles) has been reported by Cockrell (33). In addition, K⁺/H⁺ but not Na⁺/H⁺ antiport in *E. coli* vesicles is sensitive to trypsin (32). This raises the possibility that the K⁺/H⁺ component of SMP may be inactivated by proteases during preparation of the particles. Further work will be necessary to establish why K⁺/H⁺ antiport does not appear to be present in SMP.

(a) The passive swelling reactions in which cation uptake has been attributed to the action of the K⁺/H⁺ antiport show maximum activity only in Mg²⁺-depleted mitochondria at elevated pH. Both divalent cation depletion and alkaline pH produce generalized increases in the permeability of the mitochondrial membrane, with increased permeability to anions, to H⁺, and possibly to cations apparent by swelling and other criteria. It is, therefore, not possible to attribute the uptake of K⁺ under these conditions to a specific pathway (such as K⁺/H⁺ antiport) unless additional and more sensitive criteria can be made available.

(b) The inhibitors Mg²⁺, quinine, and DCCD that have been used to characterize the K⁺/H⁺ antiport all have multiple effects on the mitochondrial membrane and cannot be relied on to act as specific inhibitors of a component such as the putative antiport. The present results with DCCD emphasize that the response to this reagent is complex and will require further examination.

(c) Thus far, neither studies with mitochondria or with SMP have been able to detect K⁺-dependent H⁺ movements analogous to those seen for Na⁺/H⁺ antiport under a number of measuring conditions. In the absence of such a demonstration, an obligatory link between K⁺ and H⁺ conductance as postulated for the K⁺/H⁺ antiport remains to be established.

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SUPPLEMENTARY MATERIAL TO
K⁺/H⁺ ANTIPORT IN HEART MITOCHONDRIA

by
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MATERIALS AND METHODS

Preparation of Heart Mitochondria - Mitochondria were prepared from fresh beef hearts by the following modification of the preparation described by Jung et al. (14): About 250g of cold, well-trimmed ventricle was passed through a pre-cooled electric meat grinder (0-3") into 500 ml of an incubation medium consisting of sucrose (250 mM), Tris chloride (10 mM, pH 7.0) and EGTA (1 mM). Nagarse (50 mg) was added and the suspension stirred on ice for 15 min while maintaining the pH at 7.0 to 7.4 with Tris base. The supernate was then decanted through cheese cloth and discarded and the tissue mince washed with an additional charge of cold incubation medium. The washed tissue was suspended in 500 ml of the incubation medium containing BSA (1 mg/ml) and homogenized using a loose Teflon pestle and 50 ml homogenizing vessels. The homogenate was then diluted with an additional 700 ml of incubation medium and centrifuged for 10 min at 1750 rpm in a Sorvall GS-3 rotor. The supernate was poured through a layer of cheese cloth and centrifuged for 15 min at 8000 rpm in the GS-3 rotor. The resulting mitochondrial pellet was taken up in 125 ml of a suspending medium consisting of sucrose (250 mM), Tris chloride (5 mM, pH 7.2) and BSA (1 mg/ml) and re-isolated by centrifugation for 10 min at 12,000 rpm in a Sorvall SS-34 rotor. The fluffy layer was removed from the resulting pellets and the mitochondria taken up in the suspending medium and re-isolated by centrifugation. The resulting pellets had no visible fluffy layer and were suspended in sucrose-Tris-BSA at 25 mg protein/ml.

Divalent Cation-Depleted Heart Mitochondria - Heart mitochondria were suspended at 2 mg protein/ml in a medium of KCl (100 mM) and TES (10 mM, pH 7.0) containing A23187 (1 μ M), EDTA (0.3 mM), rotenone (3 μ g/ml) and succinate (5 mM) for 5 min at 25°. The mitochondria were re-isolated by centrifugation, washed twice in cold sucrose-TES containing BSA (2 mg/ml) and EDTA (50 μ M), and taken up in a medium of sucrose (250 mM), TES (10 mM, pH 7.0), and EDTA (50 μ M). The cation content and respiratory properties of the resulting Mg²⁺-depleted heart mitochondria are shown in Table I. These divalent cation-depleted heart mitochondria were labeled with ⁴²K⁺ by adding the label to the KCl in the above process and washing as described. The cation content of the mitochondria was established by atomic absorption spectroscopy of perchloric acid extracts. When mitochondria depleted of K⁺ as well as divalent cations were desired, the KCl was replaced in the above procedure with TEA⁺ chloride (10). The mitochondria that are depleted of K⁺ as well as divalent cations by this process show rates of state 3 respiration, respiratory control indices, and ADP:O ratios that are not significantly lower than divalent cation-depleted mitochondria that retain high levels of matrix K⁺ (see Table I). Where indicated heart mitochondria were also depleted of Mg²⁺ and K⁺ in TMA⁺ chloride (100 mM), NaCl (100 mM), or sucrose (200 mM). The cation content of these preparations was similar to that of mitochondria depleted in TEA⁺ (Table I).

Submitochondrial Particles - SMP were prepared from beef heart mitochondria by sonication in the presence of EDTA essentially as described by Spitsberg et al. (15) for liver mitochondria particles. The respiration-dependent Δ pH was followed by quinacrine fluorescence using a Perkin-Elmer MPF44 fluorimeter and incubation conditions based on those described by Rosen and Futai (16). The composition of the suspending medium is given with the experiments reported.

Analytical - The exchange and net loss of matrix ⁴²K⁺ were determined as previously described (17). Swelling and contraction was followed by the change in absorbance at 540 nm in a Gilford spectrophotometer (18). Adenine nucleotides were determined by HPLC (19) of acid extracts. State 3 respiration, respiratory control ratios, and ADP:O ratios were determined with succinate as substrate (20).

Table I - Properties of Mg²⁺-Depleted Heart Mitochondria

	Preparation		
	Untreated Mitochondria	Mg ²⁺ -Depleted in KCl	Mg ²⁺ -Depleted in TEA ⁺ Cl
Cation Content			
Mg ²⁺ (ngatom-mg ⁻¹)	45	1.3	2.1
Ca ²⁺ "	9	2.8	3.3
K ⁺ "	120	110	5
State 3 Respiration (ngatom-min⁻¹-mg⁻¹)			
	330	240	203
Resp. Control Index			
	4.4	2.8	2.5
ADP:O Ratio			
	1.9	1.8	1.7
Total Adenine Nucleotide (AMP + ADP + ATP; nmol-mg⁻¹)			
	5.0	3.4	2.9

Table II - Rates of Swelling of K⁺ and Divalent Cation-Depleted Heart Mitochondria in Nitrate Salts as a Function of pH.

Suspending Medium	Swelling Rate (Δ A ₅₄₀ -min ⁻¹)			
	Untreated Mitochondria		K ⁺ and Divalent Cation Depleted Mitochondria (TEA ⁺)	
	pH 7.1	pH 8.2	pH 7.1	pH 8.2
NH ₄ ⁺ Nitrate	0.01	0.04	0.08	0.55
" + CCCP (1.5 μ M)	0.08	0.20	0.50	1.50
K ⁺ Nitrate	<0.01	0.01	0.03	0.45
" + CCCP	0.01	0.04	0.10	0.47
" + valinomycin	--	--	1.0	1.7

Rates of swelling were estimated from records such as those of Fig 1 of the change in absorbance at 540 nm of heart mitochondria (0.5 mg ml⁻¹) suspended at 25° in a medium of the indicated salt (100 mM), TES (5 mM) at the indicated pH, EDTA (100 μ M), and rotenone (3 μ g/ml). Where indicated CCCP (1.5 μ M) or valinomycin (1.0 μ M) were also added.

Table III - DCCD Inhibits Swelling of Cation-Depleted Mitochondria in K⁺ Acetate and NH₄⁺ Nitrate but Does Not Affect ⁴²K⁺/K⁺ Exchange

	K ⁺ (nmol-mg ⁻¹)	⁴² K ⁺ /K ⁺ Exchange (nmol-mg ⁻¹ -min ⁻¹)	Swelling Rate (Δ A ₅₄₀ -min ⁻¹) K ⁺ Acetate	NH ₄ ⁺ Nitrate
No DCCD	26	26	0.116	0.336
+ DCCD	51	28 (0%)	0.017 (86%)	0.079 (76%)

Heart mitochondria were depleted of Mg²⁺ and K⁺ by treatment with A23187 and EGTA in a medium of TEA⁺ chloride (100 mM) as described in the Methods section. The cation-depleted mitochondria were resuspended (5 min at 27,000 x g), resuspended in 25 mM (42K⁺) KCl and incubated in the absence or presence of DCCD (50 nmol $\cdot$ mg⁻¹) for 2 hrs in an ice bath. The incubation medium also contained TES (10 mM, pH 7.8), rotenone and EDTA (50 μ M). After reaction with DCCD the mitochondria were washed once in sucrose (0.2M) containing TES (10 mM, pH 7.0), EDTA (50 μ M) and BSA (3 mg ml⁻¹) and re-suspended in this medium minus BSA. Swelling was then measured (conditions of Fig. 3) in K⁺ acetate (100 mM, pH 7.8) and NH₄⁺ nitrate (100 mM, pH 7.8). The rate of ⁴²K⁺/K⁺ exchange was estimated from plots of the loss of ⁴²K⁺ as a function of time (0.25, 0.5, 0.75, 1.0, 1.5, and 2.0 min) to a medium of KCl (50 mM, pH 7.0; other conditions as shown in Fig. 7). The initial K⁺ content of the washed ⁴²K⁺ labeled mitochondria was determined by atomic absorption. The values in parenthesis are the % inhibition by DCCD.

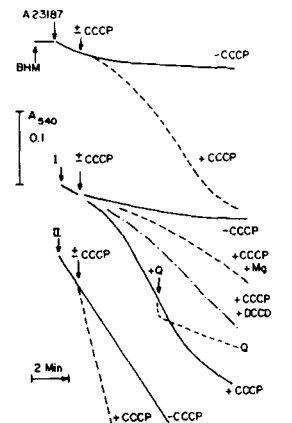


Fig. 1 - Swelling of heart mitochondria in K⁺ nitrate following depletion of divalent cations with A23187 and EDTA. Heart mitochondria were suspended at 0.5 mg protein/ml in a medium of K⁺ nitrate (100 mM), containing TES (10 mM, pH 7.1), Tris, EDTA (100 μ M) and rotenone (3 μ g/ml). Swelling was monitored at 540 nm in a Gilford spectrophotometer as previously described (18) with the temperature maintained at 25°. A23187 (1 μ M) and CCCP (1.5 μ M) were added at the indicated points. Heart mitochondria depleted of divalent cations by treatment with A23187 in a KCl medium (see Methods) or depleted of both divalent cations and K⁺ by depletion in TEA⁺ chloride (11) were suspended in the same medium. Where indicated, MgCl₂ (20 mM), CCCP (1.5 μ M), or valinomycin (0.5 μ M) was added. Also where indicated divalent cation-depleted mitochondria (11) were treated with DCCD (40 nmol-mg⁻¹ protein) for 30 min on ice prior to the swelling protocol (see 21).

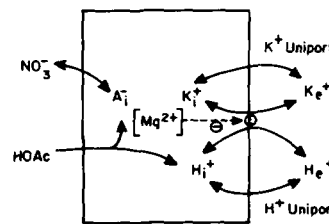


Fig. 2 - Model showing the relationship between the putative pathways for K⁺ movement in heart mitochondria. It has been proposed that the inner membrane contains a K⁺/H⁺ antiport (E) that is negatively regulated by Mg²⁺ (9,22) and H⁺ (12,29). Such a component would be activated by Mg²⁺-depletion and by elevated pH and in the active form would support passive osmotic swelling (K⁺ influx in exchange for H⁺). respiration-dependent K⁺-extrusion, or loss of K⁺ to a K⁺-free medium when an uncoupler is added to provide a pathway for H⁺ uniport. It has also been postulated (8,23) that this activated antiport promotes the passive exchange of matrix ⁴²K⁺ for external K⁺, Na⁺, or Li⁺. The available evidence also suggests that pathways for K⁺ uniport may develop in mitochondria under certain conditions (see 13, for example). An exchange of K⁺ for H⁺ could also occur via indirect electrical coupling of K⁺ and H⁺ conducting pathways, as is seen in the presence of valinomycin (K⁺ uniport) and an uncoupler (H⁺ uniport).

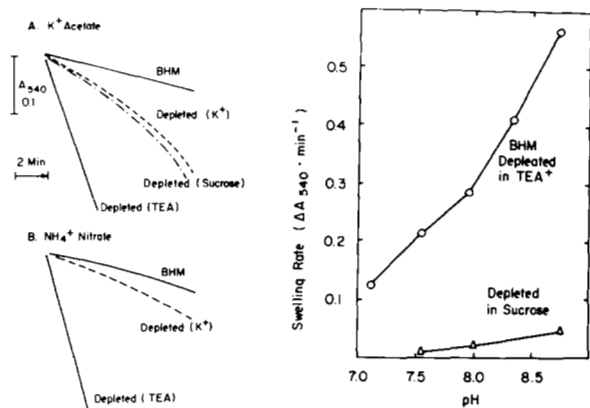


Fig. 3 - (Left) Swelling of divalent cation-depleted heart mitochondria in K^+ acetate (A) or NH_4^+ nitrate (B) at pH 7.1. The conditions were identical to those of Fig. 1 except K^+ nitrate was replaced with K^+ acetate (100 mM) or NH_4^+ nitrate (100 mM). The responses of untreated mitochondria (BHM), divalent cation-depleted mitochondria prepared in KCl (see Methods) and mitochondria depleted of both K^+ and divalent cations in either TEA^+ chloride or sucrose are shown.

Fig. 4 - (Right) Swelling in K^+ acetate of mitochondria depleted of both K^+ and divalent cations in a medium of TEA^+ chloride (see Methods) as a function of the pH of the swelling medium. The rates were estimated from records such as those of Fig. 3 with the pH adjusted as shown. TES was the buffer for pH 8.0 and below; Tricine above pH 8.0. The much lower swelling rates seen with heart mitochondria depleted of divalent cations and K^+ in a sucrose medium are also plotted.

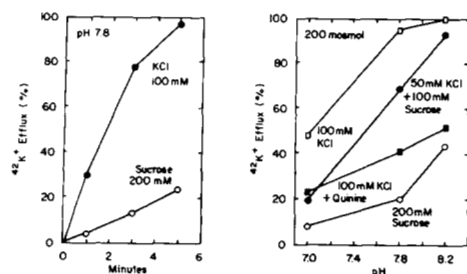


Fig. 5 - (Left) Time course of the loss of matrix $^{42}K^+$ from divalent cation-depleted heart mitochondria suspended in KCl or sucrose. Heart mitochondria were depleted of divalent cations and matrix K^+ equilibrated with $^{42}K^+$ as described in the Methods section. The $^{42}K^+$ -loaded, Mg^{2+} -depleted mitochondria were incubated (1 mg/ml) for the indicated time at 21° in a medium of either KCl (100 mM) or sucrose (200 mM) containing 75 μ M EDTA, 3 μ M rotenone and 10 mM TES adjusted to pH 7.8. The mitochondria were removed by rapid centrifugation and the $^{42}K^+$ efflux calculated as previously described (17).

Fig. 6 - (Right) Release of matrix $^{42}K^+$ from divalent cation-depleted mitochondria as a function of pH. The loss of label after 3 min to a medium of KCl (100 mM), KCl + sucrose (50 mM + 100 mM), or sucrose (200 mM) is compared at the indicated pH values. Also shown is the effect of quinine (1.0 mM) on the exchange of matrix $^{42}K^+$ with 100 mM KCl. Other conditions were identical to those of Fig. 5.

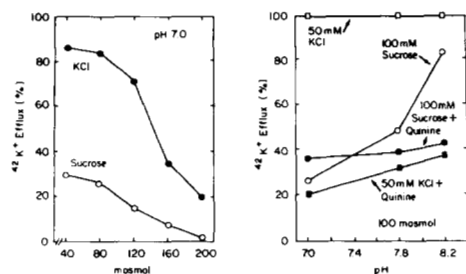


Fig. 7 - (Left) Increased passive exchange of matrix $^{42}K^+$ with external K^+ vs net loss (50) sucrose with decreasing osmolarity. Mitochondria labeled with $^{42}K^+$ and depleted of Mg^{2+} (Methods) were suspended at 1 mg/ml in either KCl or sucrose of the indicated tonicity. The pH was 7.0 (TES, 10 mM) and other conditions were identical to those of Fig. 5.

Fig. 8 - (Right) Loss of matrix $^{42}K^+$ to hypotonic KCl and sucrose media as a function of pH. The conditions were identical to those of Fig. 5 using KCl (50 mM) or sucrose (100 mM) at the indicated pH. The effect of quinine (1 mM) is also plotted.

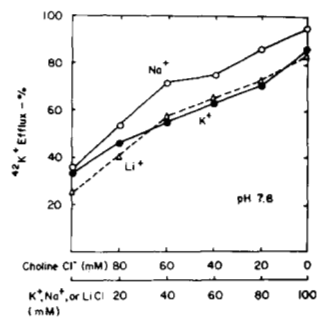


Fig. 9 - Exchange of matrix $^{42}K^+$ of divalent cation-depleted heart mitochondria against external Na^+ , K^+ , and Li^+ . Experimental conditions were as described for Fig. 5. The indicated concentrations of NaCl, KCl, or LiCl were added to choline chloride to a total salt concentration of 100 mM. The pH was 7.8 (TES, 10 mM).

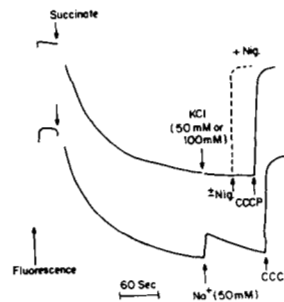


Fig. 10 - Absence of K^+/H^+ antiport in beef heart EDTA-SMP. SMP (15) were suspended at 0.8 mg/ml in a medium of choline chloride (140 mM) and Tris chloride (10 mM, pH 8.0) containing EDTA (100 μ M), oligomycin (2 μ M) and quinine (30 μ M). The fluorescence was recorded on a Perkin Elmer MPF44 fluorimeter at 25°. At the indicated point, Tris succinate (5 mM) was added to establish a Δ pH. Where indicated NaCl (50 mM), KCl (50 mM or 100 mM), nigericin (0.5 μ M) or CCCP (1.5 μ M) were added.